INCH-POUND

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14 November 1990
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10 August 1984

MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14A AMPLIFIER MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Amplifier Module described herein shall consist of this specification and the latest issue of MIL-I-85845.

REQUIREMENTS:

Item description: The Amplifier Module shall provide amplification of audio signals with gain characteristics as specified herein when used in Master Interphone Control Panel C-2645D/AIC-14A.

Item diagram: The Amplifier Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

Interface description: The Amplifier Module input and output signals will normally display the characteristics presented in Table I.

Environmental: The Amplifier Module shall meet the environmental requirements of MIL-I-85845 except the fluids requirement shall not apply.

Characteristics:

Performance characteristics:

Input characteristics:

No. 1 (NO. 1): The Amplifier Module shall have a NO. 1 audio input circuit having the following performance characteristics:

Frequency and Amplitude: The amplifier shall accept input drive signals which range from 300 to 7000 Hz. The amplifier shall accept input drive signals which range in amplitude from 0 to 1.0 Vac. The amplifier shall be capable of operating within specification when the signal input contains a dc component of up to 2 volts.

AMSC N/A FSC 5831

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TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
No. 1 (NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1.0 Vac. No dc component normally present.
No. 2 (NO. 2)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1.0 Vac. No dc component normally present.
Power (PWR)	+27.5 ± 0.5 Vdc.
Outputs:	
Headset High (HDST HI Z)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 27 Vac.
Headset Low (HDST LO Z)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 3.0 Vac.
Grounds:	
Power (PWR GND)	Power return. See MIL-I-85845.
System (SYS GND)	Signal return. See MIL-I-85845.

Input impedance: The amplifier shall have input impedance of 1200 ohms ± 10 percent.

Isolation: The isolation between signal inputs shall be not less than 20 dB.

No. 2 (NO. 2): The No. 2 input characteristics shall be as specified for Input NO. 1.

Power (PWR): The Amplifier Module shall accept and operate within the constraints of this specification when supplied with electrical power as specified in MIL-I-85845. The maximum load presented by the amplifier shall be not greater than 150 milliamperes.

Output characteristics: The Amplifier Module shall have the following output characteristics:

Headset High (HDST HI Z): The HDST HI Z output shall have the following characteristics:

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Frequency and amplitude: The HDST HI Z signal output shall exceed the input amplitude (0.06 Vac) at 1000 Hz by 44 ± 1 dB (8.5 to 10.7 Vac) when terminated with 600 ohms. The HDST HI output signal amplitude shall be in accordance with the response curves of Figure 2 when terminated in a 600 ohm resistive load. The input/output characteristics (gain) of the amplifier shall be linear within ±1 dB when the input level is varied between 0.01 and 0.10 volt.

Power: The amplifier shall be capable of producing an output of 600 milliwatts into a resistive load of 270 to 600 ohms at 1000 Hz. The amplifier shall be capable of producing an output of 1 watt into a resistive load of 601 to 1000 ohms.

Distortion: The total harmonic distortion over the frequency range of 300 to 7000 Hz and input amplitude range of 0.01 to 0.10 Vac shall be not greater than five percent under standard conditions as specified in MIL-I-85845. The total harmonic distortion over the frequency range from 300 to 7000 Hz shall be not greater than eight percent throughout the environmental envelope of MIL-E-5400 Class 2 equipment as modified by MIL-I-85845 when operated at rated power.

Phasing: The output shall lag the input by approximately 180 degrees at 1000 Hz.

Amplifier stability: The HDST HI Z output signal shall be stable and show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, or any combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

Internal noise: The amplifier shall have an output of less than 5 millivolts when operated throughout the environmental envelope as specified in MIL-E-5400 for Class 2 equipment as modified by MIL-I-85845 with the input circuit shorted to system ground.

Susceptibility to Stray Magnetic Fields: The amplifier shall be shielded or so designed that when terminated into a 600 ohm load within a magnetic field of 2 Gauss in the frequency range of 400 to 2800 Hz, the output from the amplifier shall be not greater that 5 millivolts at 400 Hz. Between 400 and 2800 Hz the output shall be not greater than a linear increase up to 50 millivolts at 2800 Hz irrespective of orientation.

Headset Low (HDST LO Z): The HDST LO Z output shall have characteristics identical to those described herein except that the output impedance shall be 7.5 ohms ± 10 percent at 1000 Hz. The amplifier shall have a 24.5 ± 1 dB gain which produces a .90 to 1.2 Vac output for a .06 Vac input. Output loads for frequency response (see Figure 2), power, amplification and distortion tests shall be 7.5 ohms ± 10 percent.

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Physical characteristics:

Dimensions: The Amplifier Module shall consist of a single encapsulated module having the dimensions of Figure 3. All outside dimensions shall be within ±0.01 inch of the nominal value shown.

Weight: Weight of the amplifier shall not be greater than 9.0 ounces.

Design and construction: The design and construction of the amplifier shall be in accordance with the requirements of MIL-I-85845.

QUALITY ASSURANCE PROVISIONS: See MIL-I-85845.

Responsibility for tests and inspections: See MIL-I-85845.

Test methods:

Examination of product: See MIL-I-85845. Units procured as spares shall not be subjected to fluids inspection (see 4.2.1.10 of MIL-I-85845).

Operational test: See MIL-I-85845.

Input impedance: Verify that the NO. 1 and NO. 2 input impedances are in accordance with requirements herein.

Gain: Verify that the amplifier gain at 1000 Hz is in accordance with requirements herein.

Frequency Response: Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements herein.

Linearity: Using a 1000 Hz input signal which is varied between 0.01 Vac and 0.10 Vac, verify that the amplifier meets the requirements herein.

Distortion: Verify the requirements herein.

Phasing: Verify the requirements herein.

Amplifier stability: Verify the requirements herein.

Internal noise: Verify the requirements herein.

Susceptibility to Stray Magnetic Fields: Verify the performance herein.

Environmental stress screening (burn-in): An environmental stress screening inspection shall be required on all Amplifier Modules. Only those modules that have successfully completed the environmental stress screening inspection shall be delivered to the procuring activity. Prior to and after each submission to the screening inspection, a room ambient full electrical performance inspection shall be performed on

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each Amplifier Module. Each Amplifier Module shall be subjected to 10 4-hour temperature cycles consisting of a 2-hour cold period at -54°C and a 2-hour hot period at +71°C. An electrical performance inspection shall be performed once daily during both hot and cold periods. Inspection shall be performed after the modules have stabilized at inspection temperatures for 30 minutes. The amplifier module shall be allowed a $\pm$3 dB gain (see Frequency and Amplitude paragraph herein) when subjected to hot/cold temperature cycling only. The thermal stress screening inspection chamber's temperature change rate shall be greater than 10°C per minute.

Special inspections: See MIL-I-85845.

Preparing Activity:
Navy - AS
(Project No. 5831-N080)

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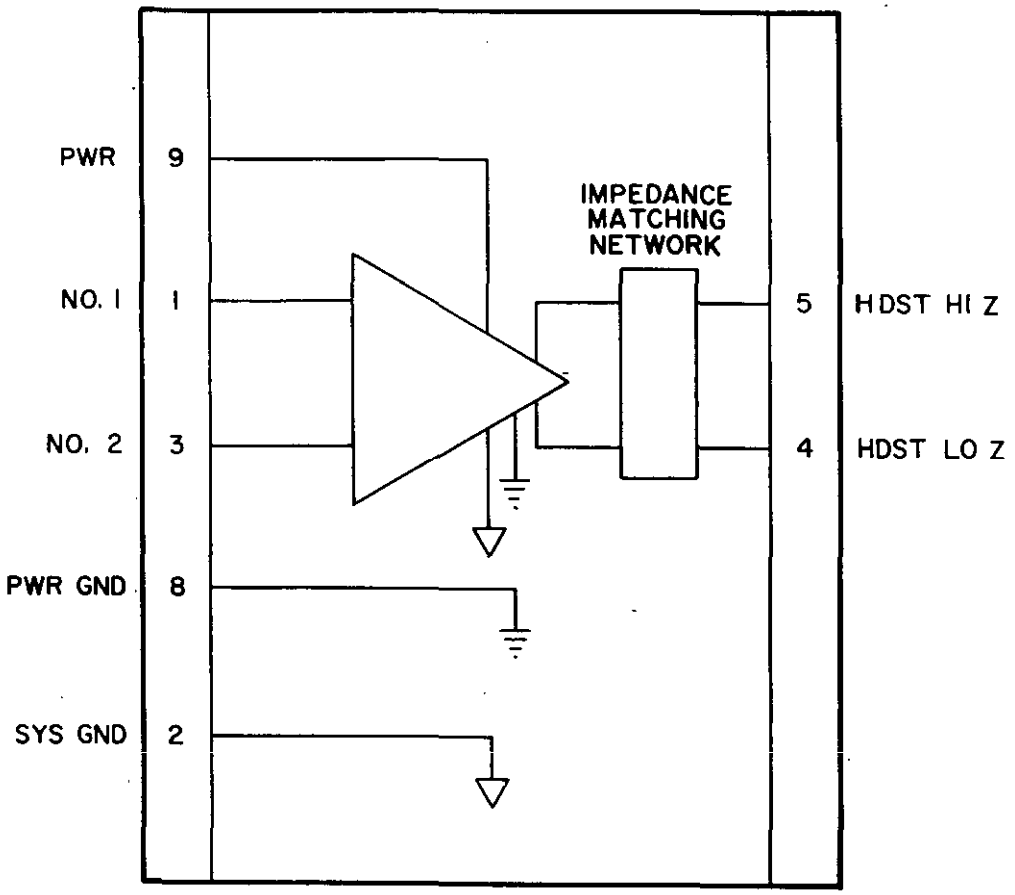
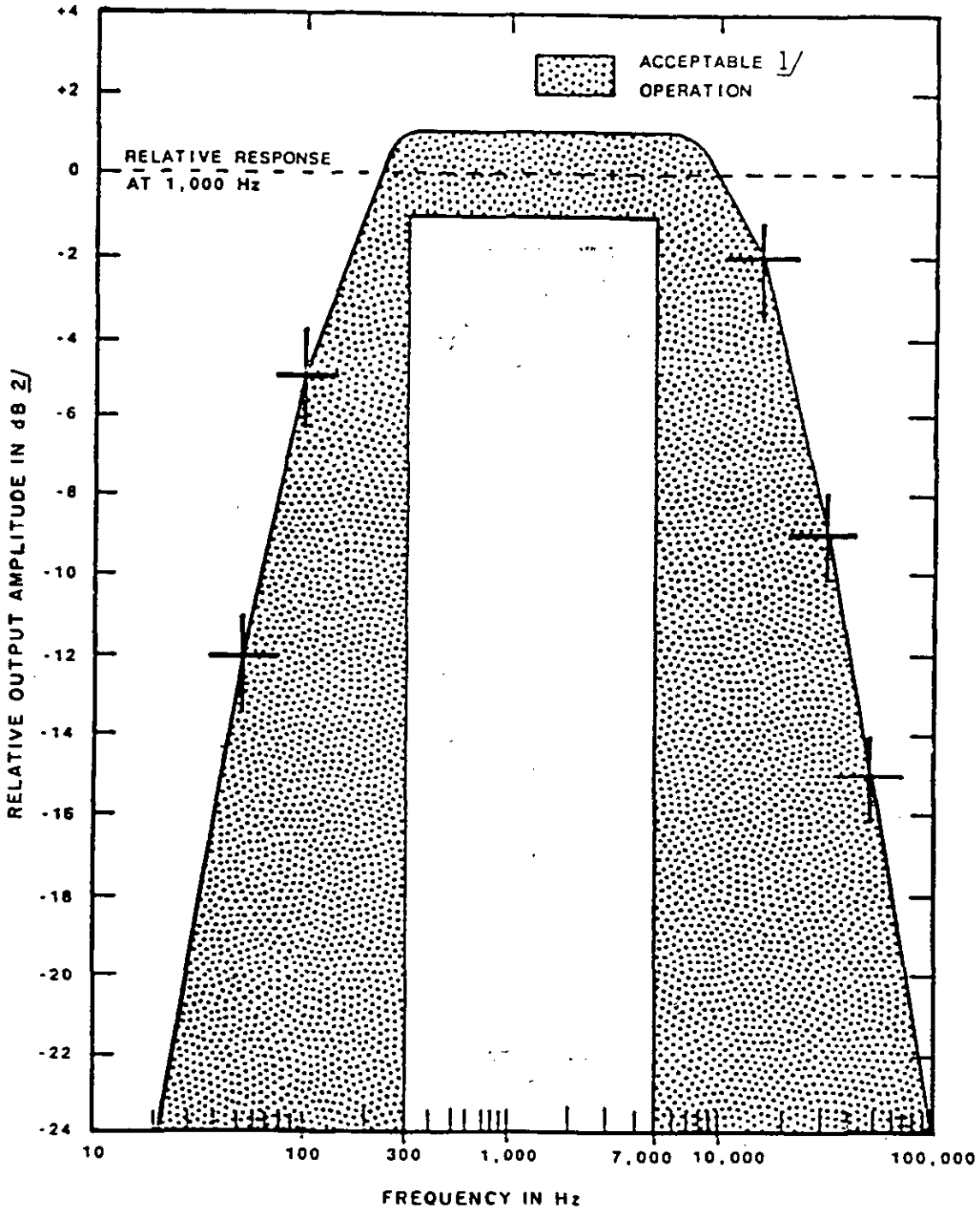


FIGURE 1. Amplifier module functional diagram.

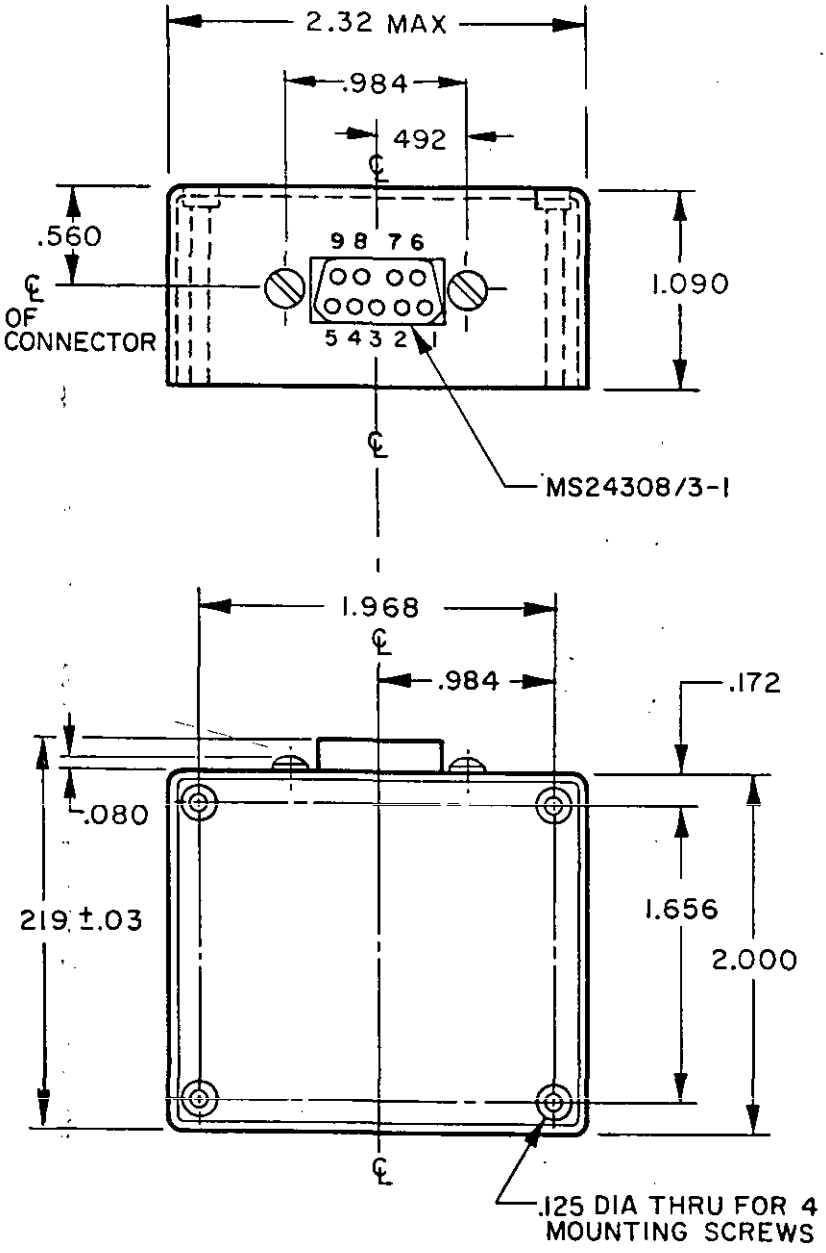
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1/ Termination into 600 ohms or 7.5 ohms.
2/ Output amplitude is measured with respect to the amplitude at 1000 Hz.

FIGURE 2. Amplifier module frequency response.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.010 inches unless otherwise specified.

FIGURE 3. Amplifier module outline drawing.

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